TOPIC 9/60

Encoder-Decoder Transformers

> The original 2017 architecture.
>
> Bringing the Reader and Writer together.

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THE REUNION

We discussed Encoders (the Readers) and Decoders (the Writers) working alone. But what happens when they collaborate?

The Best of Both Worlds:

An Encoder-Decoder model uses **both halves** in sequence. It perfectly understands the input, and perfectly generates a new output based on that understanding.

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THE TEAM ANALOGY

🤝

Two Experts

Imagine translating a book from French to English. You hire two experts.

Expert 1 (Encoder): Can only read French. Reads the book and writes down the pure "meaning" (mathematical blueprint) of the story.

Expert 2 (Decoder): Can only write English. Takes that blueprint and writes the story from scratch in English.

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HOW THEY COMMUNICATE

The Blueprint

The Encoder doesn't hand the Decoder French words. It hands over a dense, multi-dimensional matrix of numbers.

Context Vectors

This mathematical blueprint perfectly captures the grammar, tone, context, and intent of the input sentence, ready to be unwrapped by the Decoder.

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THE SECRET BRIDGE

Cross-Attention

While the Decoder is writing its output, it constantly "peeks" back at the Encoder's blueprint to make sure it's staying on track.

> This special mechanism, **Cross-Attention**, connects the two halves. The Decoder asks: "I'm generating word #4. Which parts of the original input should I focus on right now?"

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REAL WORLD EXAMPLES

Famous Models

While GPT and BERT get the most hype, Encoder-Decoder models quietly power critical enterprise tools.

Google T5

Text-to-Text Transfer Transformer. Treats every single NLP problem as a "text-in, text-out" translation.

BART (Meta)

Combines BERT's reading skills with GPT's writing skills.

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SUPERPOWER 1

TRANSLATION

The Absolute Masters of Translation.

This architecture was literally invented in 2017 to solve Google Translate. It translates French to English, Python to Java, or even Image Pixels to Text Descriptions perfectly.

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SUPERPOWER 2

Summarization

Because the Encoder deeply reads the whole text and the Decoder generates fresh text, it excels at summarization.

Give it a 10-page legal document. The Encoder builds a blueprint of the core facts. The Decoder writes a single, punchy 3-sentence summary based *only* on that blueprint.

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THE TRADE-OFF

Massive Size

Running two networks at once takes double the memory and computing power.

Overkill for Chat

If you just want an AI to chat and generate essays, a Decoder-only (GPT) is faster and cheaper.

Overkill for Search

If you just want to analyze emails for spam, an Encoder-only (BERT) is vastly more efficient.

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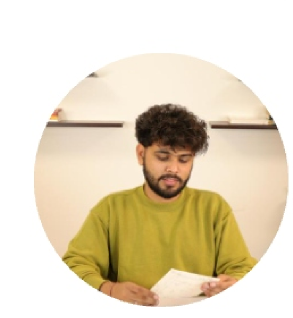

Encoder-Decoder

CONCEPT UNLOCKED

- 🤝 Combines Reader & Writer
- 🌉 Uses Cross-Attention
- 🌍 The Master of Translation
- 📝 Perfect for Summarization

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